

Tekram Technical Lead Assessment (72 Hours)

Objective

Assess whether the candidate can lead an engineering team, write production-quality code, design scalable architecture, communicate effectively, prioritize business needs, and deliver under deadlines.

Part 1 – Architecture (20%)

Design a scalable architecture for Food Delivery, Taxi, Supermarkets, Housekeeping, with future support for Pharmacy, Parcel Delivery, AI Features, Loyalty Platform, Merchant Dashboard, and Rider App Improvements. Assume 1M users, 50k DAU, 15k daily orders. Explain backend architecture, APIs, authentication, database, caching, queues, notifications, logging, monitoring, disaster recovery, CI/CD, and include diagrams.

Part 2 – Coding Challenge (25%)

Build a backend with JWT authentication (register/login), restaurant listing/search/pagination, and order creation with stock validation, delivery fee calculation, coupon support, persistence, and API responses. Bonus: Clean Architecture, Repository Pattern, Dependency Injection, Unit Tests. Deliver GitHub repository, README, setup guide, and API documentation.

Part 3 – Database Design (10%)

Design schema for Users, Restaurants, Menus, Orders, Payments, Drivers, Promotions, Ratings, Address Book, Favorites, Sessions, Notifications. Explain indexes, relationships, and scaling.

Part 4 – Debugging (10%)

Investigate an outage with 100% CPU, slow database, duplicate orders, delayed notifications, and riders unable to accept jobs. Explain investigation, root causes, immediate fixes, permanent fixes, and monitoring.

Part 5 – DevOps (10%)

Describe CI/CD, Azure architecture, Kubernetes, Redis, load balancing, secrets management, logging, backups, rollback strategy, monitoring, and deployment approach.

Part 6 – Leadership (15%)

Explain how you would manage a team of 2 Seniors, 3 Mid-level Developers, 2 Juniors, QA, and DevOps. Cover sprint planning, code reviews, documentation, mentoring, KPIs, hiring, conflict resolution, and preventing knowledge silos.

Part 7 – AI Strategy (5%)

Propose AI features for customer support, recommendations, fraud detection, demand forecasting, delivery optimization, and developer productivity.

Part 8 – Documentation (5%)

Provide architecture documentation, folder structure, deployment guide, environment variables, dependencies, API documentation, and future improvements.

Part 9 – Business Thinking (Bonus)

Explain how you would respond if orders dropped from 1,000/day to 100/day after a server outage. Include investigations, dashboards, metrics, collaboration with Growth and Marketing, first 24-hour actions, and a 30-day recovery plan.

Deliverables

GitHub repository, architecture diagrams, database schema, README, deployment guide, API documentation, leadership document, AI proposal, and technical decisions document.

Evaluation

Architecture 20, Coding 25, Database 10, Debugging 10, DevOps 10, Leadership 15, AI 5, Documentation 5. Passing score: 75/100. Minimum thresholds: Architecture 15/20, Coding 18/25, Leadership 10/15. Automatic concerns: weak architecture rationale, poor documentation, lack of testing, ignoring scalability/security, weak CI/CD knowledge, poor business thinking, or missing the deadline.